

Logic PhD Exam, May 2018

Answer six questions, at least one in each section.

Section 1

1. Provide a complete description of a formal proof system for first order logic.
2. Sketch the proof of the completeness theorem for first order logic.
3. State and prove Gödel's diagonal lemma.

Section 2

1. Sketch the proof that the Continuum Hypothesis holds in the constructible universe L .
2. Show that Zorn's lemma implies the well-ordering principle.
3. Sketch the proof of consistency of the Continuum Hypothesis using forcing.

Section 3

1. Show that every nonempty Σ_1 -definable set of natural numbers is the range of a total recursive function.
2. Sketch the proof of Friedberg–Muchnik theorem: there are two subsets of ω which are incomparable in the sense of the Turing ordering.
3. Explain what a simple set is and provide an example with a proof.